

BIOTECHNOLOGY COURSE CONTENT

Duration- 12-16 Weeks or Modular Format

OBJECTIVE:

To introduce students to the fundamentals and applications of biotechnology, covering molecular biology, genetic engineering, industrial and medical biotechnology, and emerging technologies.

MODULE 1:

Introduction to Biotechnology

- Definition, history, and scope of biotechnology
- Branches: Red (medical), Green (agricultural), White (industrial), Blue (marine) biotech
- Interdisciplinary nature: integration of biology, chemistry, and engineering
- Ethical, legal, and social issues in biotechnology

MODULE 2:

Cell and Molecular Biology

- Prokaryotic vs eukaryotic cells
- Cell structure, organelles, and functions
- DNA/RNA structure, replication, transcription, translation
- Gene expression and regulation
- Techniques: PCR, Gel electrophoresis, DNA sequencing

MODULE 3:

Genetic Engineering & Recombinant DNA Technology

- Plasmids, vectors, restriction enzymes
- Gene cloning and expression
- Transformation and selection methods
- CRISPR-Cas9 and genome editing
- Applications: transgenic organisms, gene therapy, diagnostics

MODULE 4:

Microbial and Industrial Biotechnology

- Microorganisms in biotechnology: bacteria, yeast, fungi
- Fermentation technology and bioreactors
- Enzyme production and immobilization
- Industrial applications: biofuels, bioplastics, antibiotics, alcohol
- Hands-on: Design of a simple bioreactor

MODULE 5:

Agricultural and Environmental Biotechnology

- Genetically modified crops (GMOs)
- Biofertilizers, biopesticides
- Phytoremediation and bioremediation
- Biosensors for environmental monitoring
- Case study: Bt cotton and golden rice

MODULE 6:

Medical and Pharmaceutical Biotechnology

- Vaccines (traditional and recombinant)
- Monoclonal antibodies and therapeutic proteins
- Stem cell technology and regenerative medicine
- Molecular diagnostics and biosensors
- Personalized medicine and pharmacogenomics

MODULE 7:

Bioinformatics and Emerging Trends

- Introduction to bioinformatics tools and databases (NCBI, BLAST)
- Sequence alignment and genome annotation
- Synthetic biology and systems biology
- Tissue engineering and 3D bioprinting
- Entrepreneurship in biotechnology and start-up ecosystem

MODULE 8:

Practical & Project Work

- Lab experiments: Microbial culture, DNA extraction, restriction digestion
- Industrial visit (optional)
- Mini project: Design a biotech product or business plan
- Presentation & viva

Technologies & Tools Used

- **Wet lab tools:** Micropipettes, centrifuges, spectrophotometers, PCR machines
- **Software tools:** BLAST, ClustalW, SnapGene, UniProt
- **Protocols:** DNA extraction, gel electrophoresis, plasmid isolation, enzyme assays

Course Deliverables

- Lecture slides and lab manuals
- Weekly assignments and MCQs
- Lab reports and project guide
- Final examination or evaluation
- Certificate of completion (if short course)